Exercise 2.2 SOLUTION

Q1. Find the point of intersection of the coordinate axes and the following linear functions graphically:

(i) $y = -5x + 10$

Sol: we need to find:

Y-intercept (Intersection with the *y*-axis):

Put $x = 0$

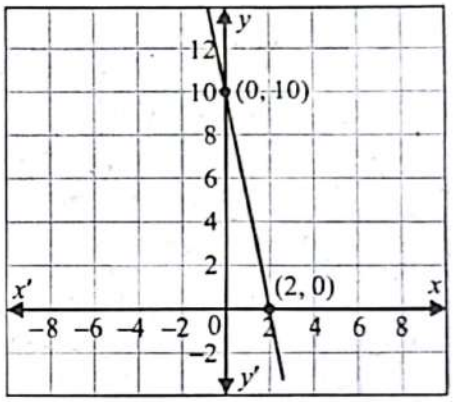
$y = -5(0) + 10 = 10 \Rightarrow$ The line intersects the *y*-axis at the point $(0,10)$.

X-intercept (Intersection with the *x*-axis):

Put $y = 0$

$0 = -5x + 10$

$\Rightarrow 5x = 10 \Rightarrow x = 2 \Rightarrow$ The line intersects the *x*-axis at the point $(2,0)$.



Graphical Summary:

- *x*-intercept: $(2,0)$
- *y*-intercept: $(0,10)$

This graph shows the line $y = -5x + 10$ intersecting:

- the *y*-axis at $(0,10)$
- the *x*-axis at $(2,0)$

These are the required points of intersection with the coordinate axes.

(ii) $y = 2x - 1$

Sol: *y*-intercept (where $x = 0$):

$y = 2(0) - 1 = -1$

$\Rightarrow$ *y*-intercept is $(0,-1)$

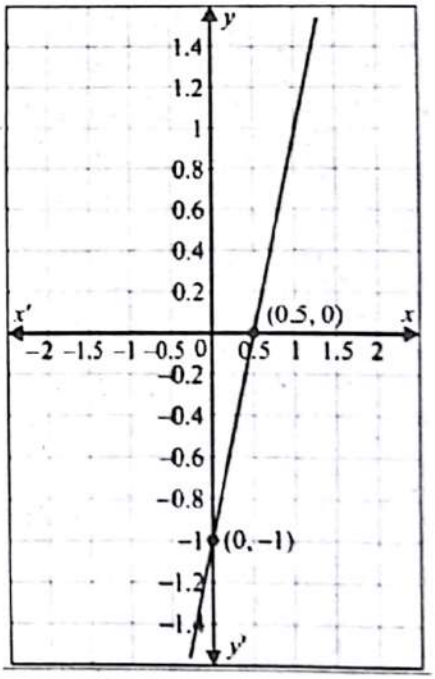
x-intercept (where $x = 0$):

$0 = 2x - 1 \Rightarrow 2x = 1$

$\Rightarrow x = \frac{1}{2} \Rightarrow$ *x*-intercept is $\left(\frac{1}{2}, 0\right)$

Now let's draw the graph of the line using these points.

Here is the graph of the line $y = 2x - 1$. It intersects:



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- the y-axis at $(0,-1)$
- the x-axis at $(0.5,0)$

(iii) $y=\frac{1}{2}x-3$

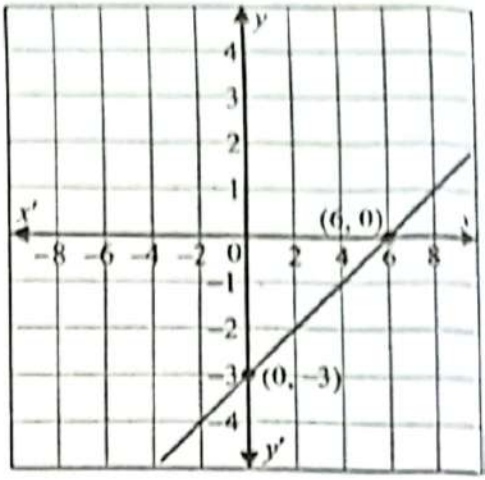
Sol: y-intercept (where $x=0$):

$y=\frac{0}{2}-3=-3 \Rightarrow y$-intercept is $(0,-3)$

x-intercept (where $y=0$):

$0=\frac{x}{2}-3 \Rightarrow \frac{x}{2}=3 \Rightarrow x=6 \Rightarrow x$-intercept is $(6,0)$

Now let's draw the graph using these points.



Here is the graph of the line $y=\frac{x}{2}-3$. It intersects:

- x-intercept at $(6,0)$ (red point)
- y-intercept at $(0,-3)$ (green point)

(iv) $y=3x+\frac{3}{2}$

Sol: y-intercept (set $x=0$):

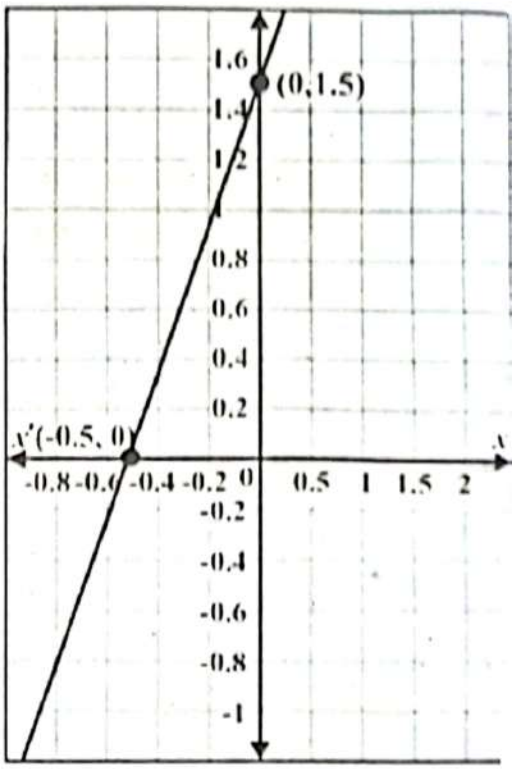
$y=3(0)+\frac{3}{2}=\frac{3}{2} \Rightarrow y$-intercept is $\left(0,\frac{3}{2}\right)=(0,1.5)$

X-intercept (set $y=0$):

$0=3x+\frac{3}{2} \quad \Rightarrow \quad 3x=\frac{-3}{2} \quad \Rightarrow \quad x=\frac{-1}{2}$

$\Rightarrow x$-intercept is $\left(\frac{-1}{2},0\right)=(-0.5,0)$.

Now let's draw the graph of this line.



Here is the graph of the line $y=3x+\frac{3}{2}$. It intersects:

i. x-intercept at $\left(\frac{-1}{2},0\right)$ ii. y-intercept at $\left(0,\frac{3}{2}\right)$

Q2: Find the pint(s) of intersection of the following functions graphically:

(i) $f(x)=2x+5, g(x)=-x+5$

Sol: For $f(x)=2x+5$

For x-intercept put $f(x)=0$

$\Rightarrow 2x+5=0 \Rightarrow x=\frac{-5}{2};\left(\frac{-5}{2},0\right)$

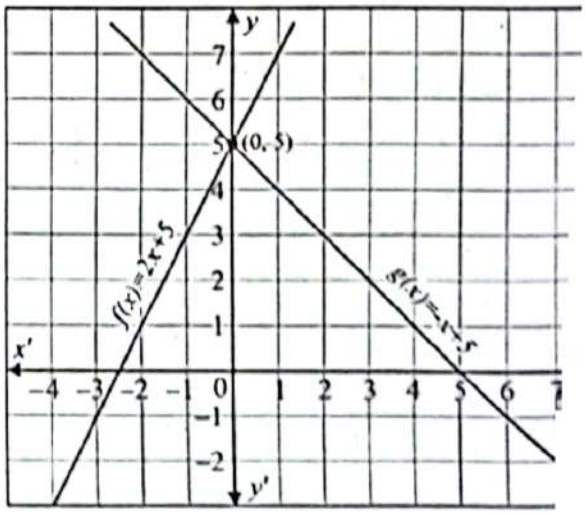
For y-intercept put $x=0$

$\Rightarrow f(x)=y=2(0)+5=5;(0,5)$

For $g(x)=-x+5$

For x-intercept put $g(x)=0$

$\Rightarrow 0=-x+5 \Rightarrow x=5;(5,0)$



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For y-intercept put $x = 0$

$\Rightarrow g(x) = y = -0 + 5 = 5; (0,5)$

Point of Intersection (Algebraically)

Set $f(x) = g(x)$

$2x+5 = -x+5 \Rightarrow 2x+x = 5-5 \Rightarrow 3x = 0 \Rightarrow x = 0$

Now substitute x=0 into either function:

$f(0) = 2(0)+5 = 5$ $\Rightarrow$ Point of intersection is $(0,5)$

Now let's plot both functions and show where they intersect.

Graph shows that two lines $f(x) = 2x+5$ and $g(x) = -x+5$ intersect at the point $(0,5)$.

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دونوں کا گراف بنانا ہے جہاں یہ دونوں گراف ملیں اس point پہ x اور y کی value کو axis پر چیک کر لینا ہے وہی دونوں کا Point of intersection

(ii) $\boldsymbol{f(x) = 3x-2, g(x) = 10-x}$

Sol: Point of intersection $f(x) = 3x-2$ and $g(x) = 10-x$ graphically:

To plot the graphs find the points of intersection with coordinate axes of the functions:

For $f(x) = 3x-2$:

x-intercept (where $f(x) = 0$):

$3x-2 = 0 \quad \Rightarrow \quad x = \frac{2}{3}$, the point: $\left(\frac{2}{3}, 0\right)$ lies on x-axes.

y-intercept (where $x = 0$):

$f(0) = 3(0)-2 = -2$, the point: $(0,-2)$ lies on y-axes.

For $g(x) = -x+10$:

x-intercept (where $g(x) = 0$):

$-x+10 = 0 \quad \Rightarrow \quad x = 10$, the point: $(10,0)$ lies on x-axes.

y-intercept (where $x = 0$):

$g(0) = -0+10 = 10$, the point: $(0,10)$ lies on y-axes.

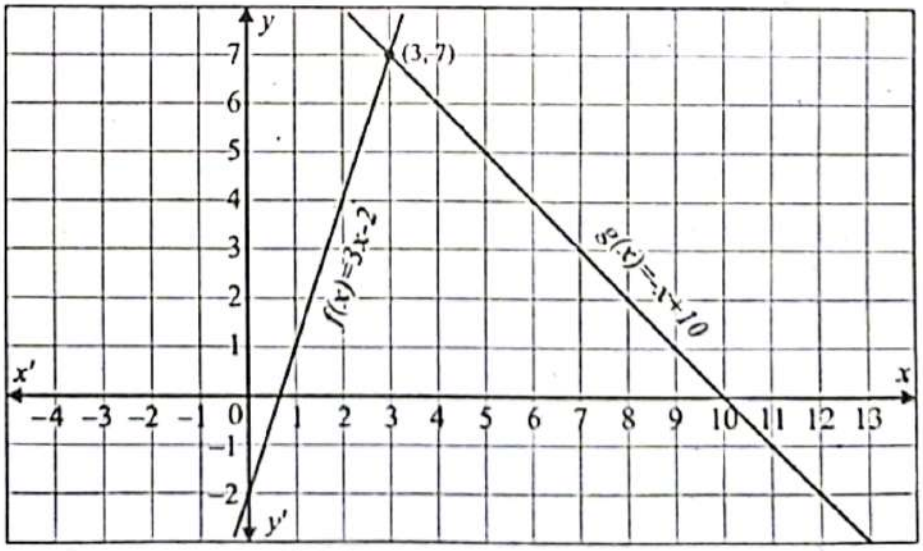
To find point of intersection of $f(x) = 3x-2$ and $g(x) = -x+10$

set $f(x) = g(x)$:

$3x-2 = -x+10 \quad \Rightarrow \quad 3x+x = 10+2 \quad \Rightarrow \quad 4x = 12 \quad \Rightarrow \quad x = 3$

Substitute $x = 3$ into either function:

$f(3) = 3(3)-2 = 9-2 = 7$ or $g(3) = 10-3 = 7$ $\Rightarrow$ Point of intersection is $(3,7)$ it is clear in graph.



Graph shows that two lines $f(x) = 3x-2$ and $g(x) = -x+10$ intersect at the point $(3,7)$.

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(iii) $f(x)=2x-4, g(x)=3x-1$

Sol: Points of intersection with coordinate axes:

For $f(x)=2x-4$:

x-intercept (Put $f(x)=0$):

$2x-4=0 \Rightarrow x=2 \Rightarrow$ Point on x-axes is $(2,0)$

y-intercept (Put $x=0$):

$f(0)=2(0)-4=-4 \Rightarrow$ Point on y-axes is $(0,-4)$

For $g(x)=3x-1$:

x-intercept (Put $g(x)=0$):

$3x-1=0 \Rightarrow x=\frac{1}{3} \Rightarrow$ Point on x-axes is $\left(\frac{1}{3},0\right)$

y-intercept (Put $x=0$):

$g(0)=3(0)-1=-1 \Rightarrow$ Point on y-axes is $(0,-1)$

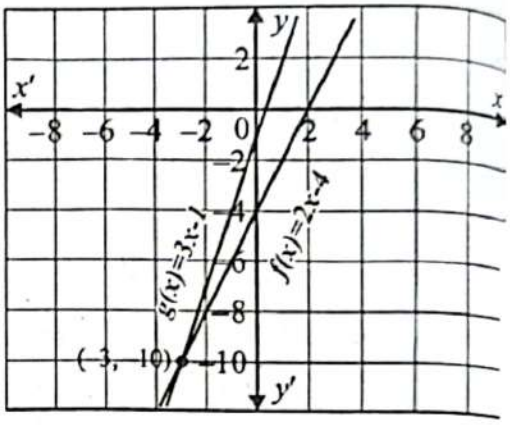
Point of intersection of $f(x)=2x-4$ and $g(x)=3x-1$:

Set $f(x)=g(x)$:

$2x-4=3x-1 \Rightarrow -4+1=3x-2x \Rightarrow -3=x$

Now putting $x=-3$ into any of the function:

$f(-3)=2(-3)-4=-6-4=-10 \Rightarrow$ Point of intersection is $(-3,-10)$



Graph shows that two lines $f(x)=2x-4$ and $g(x)=3x-1$ intersect at the point $(-3,-10)$.

(iv) $f(x)=-3x-4, g(x)=\frac{1}{2}x+3$

Sol: Points of intersection with coordinate axes:

For $f(x)=-3x-4$:

x-intercept (Put $f(x)=0$):

$-3x-4=0 \Rightarrow -3x=4 \Rightarrow x=-4/3 \Rightarrow$ Point on x-axes is $\left(-\frac{4}{3},0\right)$

y-intercept (Put $x=0$):

$f(0)=-3(0)-4=-4 \Rightarrow$ Point on y-axes is $(0,-4)$

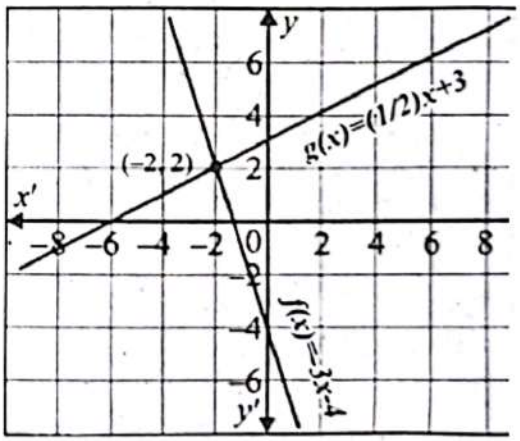
For $g(x)=\frac{x}{2}+3$:

x-intercept (Put $g(x)=0$):

$\frac{x}{2}+3=0 \Rightarrow \frac{x}{2}=-3 \Rightarrow x=-6 \Rightarrow$ Point on x-axes is $(-6,0)$

y-intercept (Put $x=0$):

$g(0)=(0)/2+3=3 \Rightarrow$ Point on y-axes is $(0,3)$



Point of intersection of $f(x)=-3x-4$ and $g(x)=\frac{x}{2}+3$:

Set them $f(x)=g(x)$:

$-3x-4=\frac{x}{2}+3$

$\Rightarrow -6x-8=x+6 \Rightarrow -6x-x=6+8 \Rightarrow -7x=14 \Rightarrow x=-2$

Now substitute $x=-2$ into any of the function:

$f(-2)=-3(-2)-4=6-4=2 \Rightarrow$ Point of intersection is $(-2,2)$

Graph shows that two lines $f(x)=-3x-4$ and $g(x)=\frac{x}{2}+3$ intersect at the point $(-2,2)$.

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(v) $f(x)=x-1, g(x)=x^2-4x+3$

Sol: Points of intersection with coordinate axes

For $f(x)=x-1$:

x-intercept (Put $f(x)=0$):

$x-1=0 \quad \Rightarrow \quad x=1 \quad \Rightarrow$ the point on x-axes: $(1,0)$

y-intercept (Put $x=0$):

$f(0)=0-1=-1 \quad \Rightarrow$ the point on y-axes: $(0,-1)$

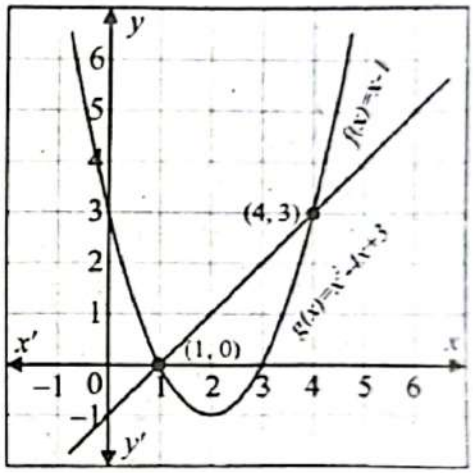
For $g(x)=x^2-4x+3$:

x-intercepts (Put $g(x)=0$):

$x^2-4x+3=0 \Rightarrow (x-1)(x-3)=0 \Rightarrow x=1, x=3 \Rightarrow$ The points are: $(1,0)$ and $(3,0)$

y-intercept (Put $x=0$):

$g(0)=(0)^2-4(0)+3=3 \Rightarrow$ the point: $(3,0)$



Points of intersection of $f(x)=x-1$ and $g(x)=x^2-4x+3$

Set the functions equal:

$x-1=x^2-4x+3 \Rightarrow 0=x^2-5x+4 \Rightarrow x^2-x-4x+4=0$

$\Rightarrow x(x-1)-4(x-1)=0 \Rightarrow (x-4)(x-1)=0 \Rightarrow x=1, x=4$

Now find corresponding y values:

For $x=1$:

$f(1)=1-1=0 \quad \Rightarrow$ the point is $(1,0)$

For $x=4$:

$f(4)=4-1=3 \Rightarrow (4,3)$

Points of intersection: $(1,0)$ and $(4,3)$.

Graph shows that two lines $f(x)=x-1$ and $g(x)=x^2-4x+3$ intersect at the points $(1,0)$ and $(4,3)$

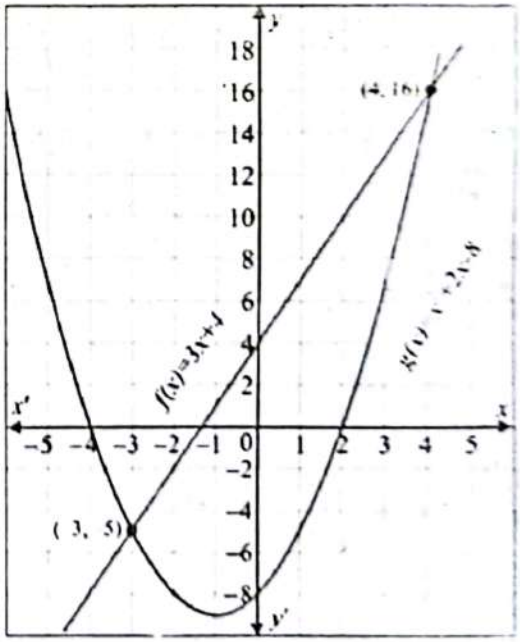
(vi) $f(x)=3x+4, g(x)=x^2+2x-8$

Sol: Points of Intersection with Coordinate Axes

For $f(x)=3x+4$:

x-intercept (Put $f(x)=0$):

$3x+4=0 \quad \Rightarrow \quad x=-4/3 \quad \Rightarrow$ The point on x-axes: $\left(-\frac{4}{3},0\right)$



y-intercept (Put $x=0$):

$f(0)=3(0)+4=4 \quad \Rightarrow \quad$ The point on y-axes: $(0,4)$

For $g(x)=x^2+2x-8$:

x-intercepts (Put $g(x)=0$):

$x^2+2x-8=0 \Rightarrow (x+4)(x-2)=0 \Rightarrow x=-4, x=2$

The points on x-axes are: $(-4,0)$ and $(2,0)$

y-intercept (Put $x=0$):

$g(0)=(0)^2+2(0)-8=-8 \Rightarrow$ The point on y-axes: $(0,-8)$

Points of Intersection of $f(x)=3x+4$ and $g(x)=x^2+2x-8$

Take $f(x)=g(x)$:

$3x+4=x^2+2x-8 \quad \Rightarrow \quad 0=x^2-x-12 \quad \Rightarrow \quad (x-4)(x+3)=0 \quad \Rightarrow \quad x=4, x=-3$

Now find y values for both:

For $x=4$:

$f(4)=3(4)+4=12+4=16 \quad \Rightarrow$ the point is $(4,16)$:

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For x=−3:

$f(-3)=3(-3)+4=-9+4=-5 \quad \Rightarrow$ the other point is $(-3,-5)$:

From graph the Intersection points of $f(x)$ and $g(x)$ are $(4,16)$ and $(-3,-5)$:

(vii) $f(x)=-2x-1, g(x)=x^2-4x$

Sol: For $f(x)=-2x-1$

x-intercept (Put $f(x)=0$):

$$-2x-1=0\Rightarrow -2x=1\Rightarrow x=\frac{-1}{2};\ \left(-\frac{1}{2},0\right)$$

y-intercept (Put $x=0$):

$f(x)=y=-2(0)-1\Rightarrow y=-1;\ (0,-1)$

For $g(x)=x^2-4x$:

x-intercepts (Put $g(x)=0$):

$x^2-4x=0\Rightarrow x(x-4)=0\Rightarrow x=0\ \&\ x=4$

We get $(0,0),(4,0)$

y-intercept (Put $x=0$):

$g(x)=y=(0)^2-4(0)=0\Rightarrow(0,0)$

For Intersection of $f(x)$ and $g(x)$

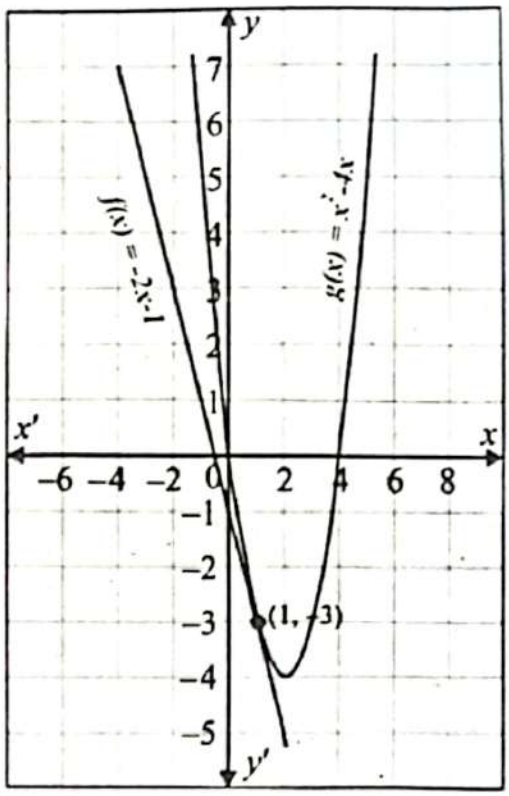
Take $f(x)=g(x)$

$-2x-1=x^2-4x\Rightarrow x^2-4x+2x+1=0$

$x^2-2x+1=0\Rightarrow(x-1)^2=0\Rightarrow x-1=0\Rightarrow x=1$

Put $x=1$ in $f(x)$ we get $f(1)=-2(1)-1=-3$

Hence point of intersection is $(1,-3)$



(viii) $f(x)=-x^2-3x+2, g(x)=x+6$

Sol: For $f(x)=-x^2-3x+2$

x-intercept (Put $f(x)=0$):

$-x^2-3x+2=0\Rightarrow x^2+3x-2=0$

$$x^2+3x-2=0\Rightarrow x=\frac{-3\pm\sqrt{(3)^2-4(1)(-2)}}{2(1)}$$

$$\Rightarrow x=\frac{-3\pm\sqrt{9+8}}{2}=\frac{-3\pm\sqrt{17}}{2}=\frac{-3\pm4.123}{2}$$

$$x=\frac{-3+4.123}{2}\quad,\quad x=\frac{-3-4.123}{2}$$

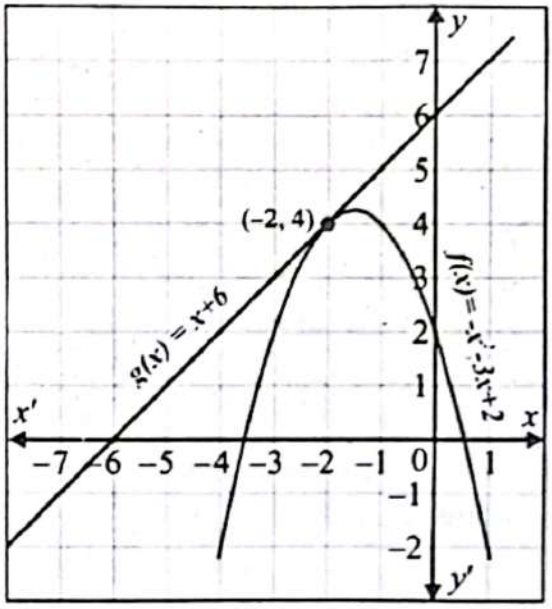
$$x=\frac{1.123}{2}\quad,\quad x=\frac{-7.123}{2}$$

$x=0.56\quad,\quad x=-3.56$

We get $(0.56,0)$ and $(-3.56,0)$

y-intercept (Put $x=0$):

$f(x)=y=-(0)^2-3(0)+2=2;(0,2)$



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For $g(x) = x + 6$:

x-intercepts (Put $g(x) = 0$):

$x + 6 = 0 \Rightarrow x = -6 \; ; (-6, 0)$

y-intercept (Put $x = 0$):

$g(x) = y = 0 + 6 = 6 \; ; (0, 6)$

For Intersection of $f(x)$ and $g(x)$

Take $f(x) = g(x)$

$-x^2 - 3x + 2 = x + 6 \Rightarrow x^2 + 3x - 2 + x + 6 = 0$

$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = x + 2 = 0 \Rightarrow x = -2$

Put $x = -2 \Rightarrow f(-2) = -(-2)^2 - 3(-2) + 2 = -4 + 6 + 2 = 4$

$(-2, 4)$ Is point of intersection

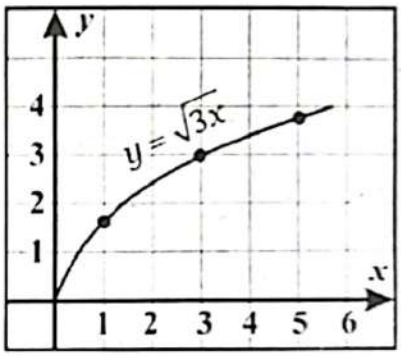
Q3. Graph the following functions:

(i) $y = \sqrt{3x}$

x	0	1	3	5
y	0	1.73	3	3.87

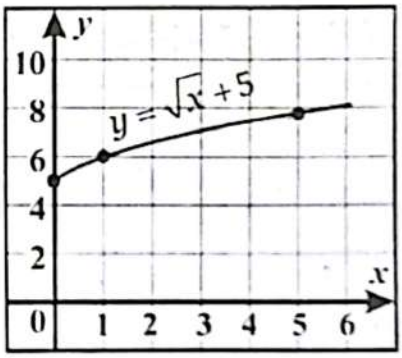
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X کی value جمع کی اس لیے لی گئی ہیں کہ x کے اوپر جذر ہے اور جذر کے اندر منفی نہیں لگاتے اس سے *Iota i* آجاتا ہے



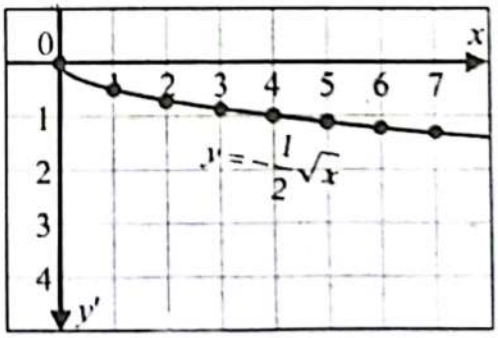
(ii) $y = \sqrt{x} + 5$

x	0	1	3	5
y	5	6	6.73	7.23



(iii) $y = -\frac{1}{2}\sqrt{x}$

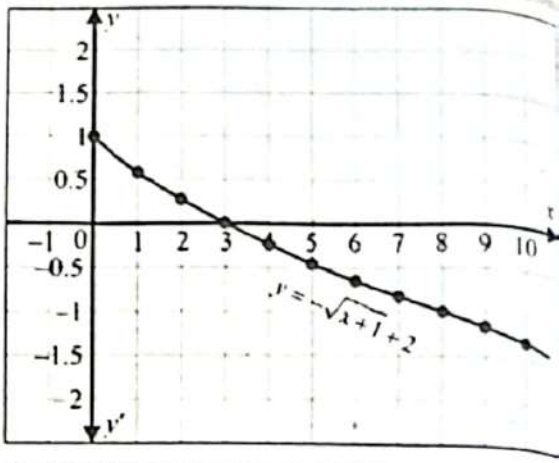
x	0	1	3	5
y	0	−0.5	−0.86	−1.12



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(iv) $y = -\sqrt{x+1} + 2$

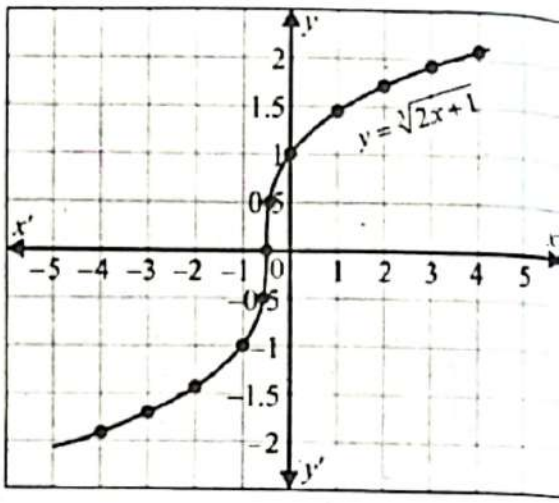
x	0	1	3	5
y	1	0.59	0	−0.45



(v) $y = \sqrt[3]{2x+1}$

x	0	1	3	5
y	1	1.44	1.91	2.22

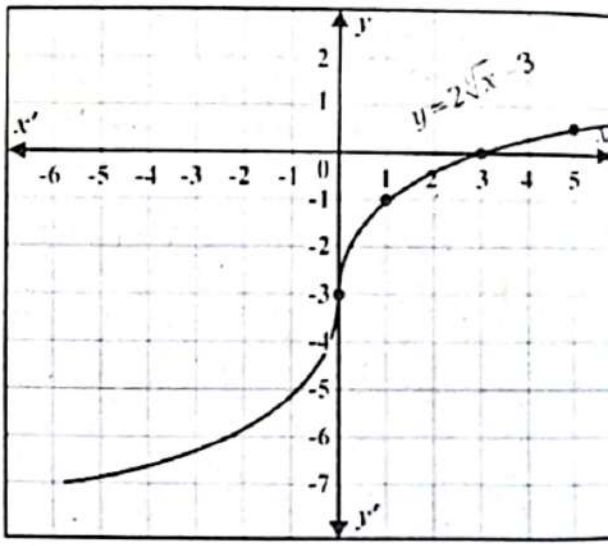
Graph of $y = \sqrt[3]{2x+1}$



(vi) $y = 2\sqrt[3]{x} - 3$

X	0	1	3	5
Y	−3	−1	−0.16	0.42

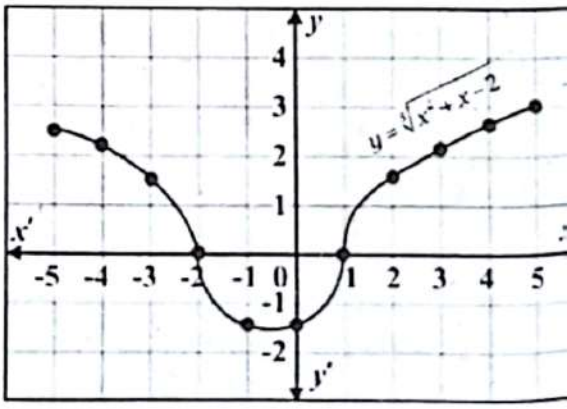
Graph of $y = 2\sqrt[3]{x} - 3$



(vii) $y = \sqrt[3]{x^2 + x - 2}$

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y	2.62	2.15	1.59	0	-1.26	-1.26	0	1.59	2.15	2.62	3.04

Graph of $y = \sqrt[3]{x^2 + x - 2}$



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Q4. A building's height over time is modeled by $H(t) = 100 + 20t$ which is in meters and t is the time in months. The height of growing tree nearby is given by $T(t) = 50 + 10t + t^2$.

(i) At what time will the building and tree have the same height?

(ii) What will that height?

Sketch the graphs of both functions and determine the time when the tree will overtake the height of the building.

Sol: We are given two functions for height over time t (in months):

Building: $H(t) = 100 + 20t$

Tree: $T(t) = 50 + 10t + t^2$

We are to find:

(i) **When the building and tree have the same height:**

Set the two height functions equal:

$100 + 20t = 50 + 10t + t^2$

Rearranging:

$100 + 20t = t^2 + 10t + 50 \quad \Rightarrow \quad 0 = t^2 - 10t - 50$

This is a quadratic equation:

$t^2 - 10t - 50 = 0$

Use the quadratic formula:

$$t = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(1)(-50)}}{2(1)}$$

$$\Rightarrow t = \frac{10 \pm \sqrt{300}}{2} = \frac{10 \pm \sqrt{300}}{2} = 5 \pm 5\sqrt{3}$$

So, the two times are:

$t = 5 + 5\sqrt{3}$ and $t = 5 - 5\sqrt{3}$. Since t is time (months), and must be positive, discard the negative one:

$t = 5 + 5\sqrt{3} \approx 5 + 5(1.732) = 5 + 8.66 = 13.66$ months ≈ 14 months (approximately)

(ii) **What is that height?**

Substitute $t = 5 + 5\sqrt{3}$ into either function (let's use building height):

$$H(t) = 100 + 20t = 100 + 20\left(5 + 5\sqrt{3}\right) = 100 + 100 + 100\sqrt{3} = 200 + 100\sqrt{3}$$

$$\Rightarrow \quad H(t) \approx 200 + 100(1.732) = 200 + 173.2 = 373.2 \text{ meters (approx.)}$$

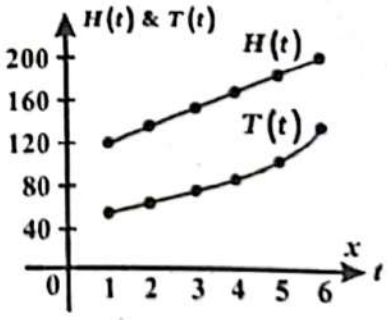
Graph of the functions: Building height: $H(t) = 100 + 20t$ and Tree height: $T(t) = 50 + 10t + t^2$.

Building Height $H(t) = 100 + 20t$

t	1	2	3	4	5
$H(t)$	120	140	160	180	200

For Tree $T(t) = 50 + 10t + t^2$

t	1	2	3	4	5
$T(t)$	61	74	89	106	125



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Q5: A radioactive substance has half-life (*h*) of 2 years. If the initial quantity Q_0 is 200 grams and the exponential decay function is $Q(t) = Q_0\left(\frac{1}{2}\right)^{\frac{t}{h}}$, then find the remaining quantity after 6 years graphically.

Sol: We are given:

- Half-life: $h = 2$ years
- Initial quantity: $Q_0 = 200$ grams
- Time: t=6 years
- Decay model:

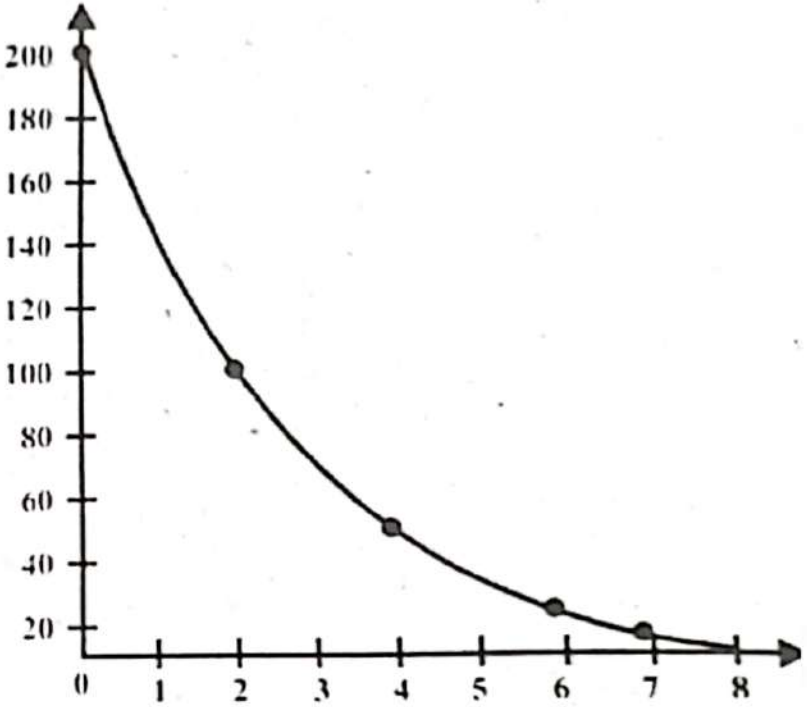
$$Q(t) = Q_0 \cdot \left(\frac{1}{2}\right)^{\frac{t}{h}}$$

Step-by-step calculation:

$$Q(6) = 200 \cdot \left(\frac{1}{2}\right)^{\frac{6}{2}} = 200 \cdot \left(\frac{1}{2}\right)^{3} = 200 \cdot \frac{1}{8} = 25 \text{ Grams.}$$

Graph of $Q = 200 \cdot \left(\frac{1}{2}\right)^{\frac{t}{2}}$

t=x	0	2	4	6	7
Q=y	200	100	50	25	17.68



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